Shyam Sundar Jayakumar

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EDUCATION

Master of Science, Aerospace Engineering **Aug 2025 - Present**

University of Illinois Urbana Champaign **GPA – 3.74/4.0**

- Related coursework: Multiphase flows, Viscous Flows, Modal Analysis of Fluid flows, Instability and Transitions, Advanced review of Math and Finite Element Analysis

Bachelor of Engineering, Aeronautical Engineering **Sep 2020 - Apr 2024**

Kumaraguru College of Technology, Coimbatore, Tamil Nadu **GPA – 3.7/4.0**

TECHNICAL SKILLS

CAD & CFD: OpenFOAM, ANSYS Fluent, ANSYS CFX, CATIA, SolidWorks, ParaView, CFD-Post

Programming: Python, MATLAB, C, C++

Core Areas: Aerodynamics, Fluid Mechanics, Turbulence Modeling, Finite Element Analysis and Design and Analysis

WORK AND RESEARCH EXPERIENCE

CFD Research Engineer - ExaSlate, India **Nov 2024 – Aug 2025**

- Developed and implemented CFD simulations using Open FOAM and ANSYS Fluent to analyse steady and transient flows and complex turbulent flow behaviour in engineering systems including grid generation and post processing
- Developed a solver to study the flow behaviour of fuel rod bundle analysis, to study velocity distribution and pressure drop characteristics and compared turbulence models (k-ε, k-ω SST, EBRSM) for CFD validation between commercial products and solver for studying shear flows and near-wall regions
- Validated simulation results against analytical models and literature, reduced simulation error vs literature benchmarks from 7% to 3.4% by evaluating the peak velocity fields in the fuel rod bundle channels.
- Designed and analysed a rear and front wing of a FSAE race car using OpenFOAM steady state solver to optimise the design of the wings to improve its downforce production up to 63% when compared with no wings attached. Contributed to peer-reviewed publications, demonstrating strong research capability and technical communication

Tool Design Engineer – TATA Advanced Systems limited, Bangalore, India **Jun 2024 – Oct 2024**

- Designed and developed mechanical components using CATIA, focusing on structural integrity and manufacturability
- Created detailed 3D models and engineering drawings to approve parts to be manufactured, ensuring compliance with ASME standards and GD&T.
- Conducted basic structural and performance analysis to validate design feasibility before prototyping.
- Modified and analysed parts of a trolley to move the Rudder of an aircraft and manufactured it to move the part across the plant for painting and post processing and reduced the manufacturing cost by 4%.
- Used Shape design and Sheet metal designing to design and modify the tools to manufacture the sections nose cone and fuselage of an aircraft to prevent the manufacturing of new tools and increase in lean production.

CONFERENCE PUBLICATIONS:

- Design and Multi-Perspective Investigations on Aeroacoustic Noise Reduction Technologies for Anti-Drone Propeller. **ASME Gas Turbine India Conference, 2023**
  DOI: https://doi.org/10.1115/GTINDIA2023-117639
- Numerical Investigation of Flow Field Around a FSAE race car body with and without the Influence of Wings Installation. **SAE Technical Papers, 2024**
  DOI: http://doi.org/10.4271/2024-01-5247

- Structural Integrity Investigation on Advanced 4-Bladed H-Darrieus VAWT for Various Lightweight Materials through Fluid–Structure Interaction Analyses. **AIAA SciTech Conference, 2025**
  DOI: https://doi.org/10.2514/6.2025-1812

JOURNAL PUBLICATIONS:

- Design, Control, Aerodynamic Performance, and Structural Integrity Investigations of Compact Ducted Drone with Co-axial Propeller for High Altitude Surveillance. **Scientific Reports, Springer Nature, 2024**
  DOI: https://doi.org/10.1038/s41598-024-54174-x
- Design and Multi-Perspective Investigations on the Aerodynamic Performance Factors of Conventional and Advanced UAV Micro Gas Turbine Engine Nozzle through Validated CFD. **International Journal of Fluid Mechanics Research, 2024**
  DOI: https://doi.org/10.1615/InterJFluidMechRes.2024051464
- Multi-Perspective Investigations of Aerosol's Nonlinear Impact on Unmanned Aerial Vehicle for Air Pollution Control Applications under Various Aerosol Environments. **Aerosol Science and Engineering, 2024**
  DOI: https://doi.org/10.1007/s41810-024-00219-7

PROJECTS

Modal decomposition of separated turbulent boundary layer **Jan 2026 - Present**

- Applied modal decomposition techniques (POD, SPOD, DMD) to transient CFD/PIV datasets of unsteady turbulent flows, extracting dominant coherent structures and identifying frequency-resolved dynamics in separated turbulent boundary layer flows using MATLAB.
- Developed a reconstruction framework using energy, spectral and dynamics-based methods evaluating backflow percentage for data-driven flow analysis and reduced-order modelling.
- Quantified flow separation dynamics using a backflow percentage metric, evaluated both as a time-resolved signal and spatial profiles along wall-normal directions.
- Assessed reconstruction accuracy using RMSE and correlation metrics, demonstrating improved physical interpretability with SPOD compared to POD and DMD on approximating shear layer and separation dynamics.

Numerical and computational investigations of Tilt Wing UAV and its control dynamics **Sep 2023 – Apr 2024**

- Designed and developed an efficient fixed wing UAV capable of performing vertical take-off and landing using CATIA, ANSYS and MATLAB.
- Estimated the design parameters of the UAV such as wingspan, propeller dimension as required for mission profile requirements.
- Computational analysis of the components to validate their outcomes such as thrust and lift production. Validated the CFD results with existing data on the thrust of the motor when operated with respective RPMs with correlation of 95%.
- Component selection was carried out to select the appropriate components required to fulfil the requirements of the mission
- Designed an altitude controller to ensure the vertical stability of the UAV using MATLAB and tuned it to meet the linear profile of altitude increment from the controller.

Tilting and folding mechanism for VTOL UAV **Sep 2023 - Apr 2024**

- Designed and developed a UAV prototype configuration compatible with efficient flight and compact storage
- Component selection for the servo motors, which need to produce enough torque to tilt and fold the rotors and wings respectively.
- Estimated the gear dimensions to produce a smooth transition between the initial stage corresponding to compact storage capability and final stage of being able to take off.
- Assembly of gears with the rotors and sample wings to validate the component selection was performed.
- Performed structural integrity tests through CATIA and ANSYS to validate the strength of the gear mechanism and used the outcomes to select appropriate materials to manufacture the prototype.
